

INDIA.RUNS

Build what next India runs on

Team Name : 2+

Team Leader Name : Anmol Raj

Problem Statement : Senior AI Engineer - Founding Team Candidate Ranking

Team Details & Introduction

PRIMARY CONTACT	DEVELOPMENT FOCUS	SUBMISSION COMPLIANCE
■ Name: Anmol Raj ■ Email: anmolraj499@gmail.com ■ Phone: +91-6204223789 ■ Role: ML Engineer & Pipeline Developer	■ 1. Offline Focus: Engineering highly efficient algorithms designed to process massive candidate datasets locally without network latency. ■ 2. Hardware Optimizations: Stream-reading data to prevent out-of-memory errors on typical recruiter computer hardware. ■ 3. Hybrid Search: Combining deep semantic transformers with keyword indexes to get explainable shortlists.	■ 1. Team ID Match: Official Team ID is 2+, matching our registered team name for consistency across portals. ■ 2. Metadata Verified: Metadata template fully populated and verified to match official hackathon requirements. ■ 3. Output Conformity: All output rankings passed formatting validation. Excel and CSV variants checked.

Problem Statement

THE RECRUITMENT BOTTLENECK

- **1. Volume Overload:**
Reviewing 100k+ candidate profiles manually is a massive time sink. Recruiters struggle to find qualified profiles in reasonable timelines.
- **2. Keyword Limitations:**
Traditional applicant tracking systems (ATS) rely on exact keyword matching, filtering out strong candidates who use adjacent terminology.
- **3. Semantic Blindness:**
Unstructured text summaries in CVs contain massive context that standard relational database filters destroy.

DATA INTEGRITY HAZARDS

- **1. Resume Inflation:**
Hackathon datasets contain trap profiles (honeypots) that claim impossible timeline metrics or fraudulent skill proficiencies.
- **2. Timeline Discrepancies:**
Profiles claiming years of experience that exceed their actual years since graduation bypass basic filters.
- **3. Notice Period Agility:**
Long notice periods lead to severe recruiter drop-offs. Traditional sorting fails to integrate notice constraints dynamically.

Proposed Solution

DECOUPLED TWO-STAGE FUNNEL	FUSION OF BEHAVIORAL SIGNALS	RECRUITER EXPLAINABILITY
■ 1. Data Integrity Gate: Programmatic filters stream-read profiles to neutralize fraudulent honeypots before any model scoring. ■ 2. Sparse Recall: A fast BM25 lexical index filters 100k down to the top 2,000 candidates in 10 seconds. ■ 3. Semantic Re-ranking: Dense vector encoders and Cross-Encoders score semantic fit locally on CPU within minutes.	■ 1. Signal Integration: Integrates notice period constraints, relocation status, and recruiter response rates directly into the ranking equation. ■ 2. Notice Agility: Immediate availability triggers ranking bonuses; standard 90-day locks receive minor ranking penalties. ■ 3. Activity Weighting: Weights scores based on candidates' active platform engagement and responsiveness metrics.	■ 1. Factual Reasonings: Generates direct, factual, non-hallucinated explanations for why a candidate is shortlisted. ■ 2. Interactive UI Sandbox: Gradio dashboard lets recruiters adjust Job Description in real-time and inspect candidates' score breakdowns. ■ 3. Verified Outputs: Enforces strict monotonic ranking format compliance, making the output instantly reliable.

System Architecture

STAGE 1 & 2: DUAL RETRIEVAL

- **1. Streaming Honeypot Filtering:**
Stream-reads the JSONL candidate dataset record-by-record, keeping memory footprint under 2GB RAM.
- **2. Temporal Integrity Checks:**
Evaluates total years of experience, skill durations, and job title coherence to filter out 35k trap candidates.
- **3. BM25 Sparse Search Funnel:**
Uses custom BM25 index over JD keywords to rapidly funnel pool from 64k down to 2,000 in ~10 seconds.

STAGE 3: DENSE EMBEDDING

- **1. Local Dense Encoding:**
Computes 384-dimensional dense vectors for top 2,000 summaries locally on CPU using SBERT all-MiniLM-L6-v2.
- **2. Semantic Overlap Mapping:**
Generates cosine similarity matrix against expanded Job Description targets, avoiding simple keyword match constraints.
- **3. Platform Signal Modulation:**
Fuses Redrob signals (immediate availability bonus, relocation willing, activity) directly into the score math.

STAGE 4: DEEP RE-RANKING

- **1. Cross-Encoder Joint Attention:**
Loads ms-marco-MiniLM-L-6-v2 locally on CPU to jointly evaluate JD-profile context on the top 200 candidates.
- **2. Joint Relevance Calibration:**
Generates a sigmoid-scaled relevance rating capturing deep contextual alignments that bi-encoders miss.
- **3. Factual Reasonings Output:**
Programmatically compiles non-templated rationales detailing years of experience, skills, and notice period constraints.

Core Algorithms & Logic

HONEYPOT IDENTIFICATION ENGINE

■ 1. Experience-Graduation Audit:

Flagged: $\text{Experience} > (\text{2026} - \text{first_graduation_year}) + 2$. This filters profiles containing impossible career timelines.

■ 2. Skill Durations Audit:

Flagged: 4+ skills claiming 'expert' or 'advanced' proficiency but specifying exactly 0 duration months.

■ 3. Career Dates Verification:

Flagged: Job duration in months exceeds the dates difference (start_date to end_date) by more than 12 months.

HEURISTIC SCORING MATH

■ 1. Notice Period Modifiers:

Notice period agility: +5% bonus for ≤ 15 days notice; -15% penalty for ≥ 90 days locked periods.

■ 2. Experience Fit Curve:

Seniority Fit: Underqualified candidates face a steep exponential penalty. Overqualified profiles are decayed slowly.

■ 3. Geographic Modulations:

Location: 1.0 multiplier for target hubs (Pune, Noida, Delhi NCR). 0.95 for relocation willing. 0.7 for non-willing.

Technologies Used

DEEP LEARNING & NLP WORKloads

- **PyTorch & Transformers:**
Provides the core tensor computation framework, optimized for local CPU execution.
- **SentenceTransformers (all-MiniLM-L6-v2):**
Computes fast dense text embedding vectors for candidates summary profiles.
- **CrossEncoder (ms-marco-MiniLM-L-6-v2):**
Computes joint attention maps across candidates and JD requirements for fine-grained re-ranking.

DASHBOARD & INTERACTION ENGINE

- **Gradio Web App:**
Powers our interactive, sandboxed web dashboard. Allows real-time JD changes and live shortlist monitoring.
- **Pandas & NumPy:**
Manages index arrays, matrix similarity multiplications, and candidate data structures.
- **OpenPyXL (XLSX Generator):**
Compiles the final validated candidate shortlist into spreadsheet format.

LOCAL ARCHITECTURE CHOICES

- **Zero-Network local Cache:**
Pre-downloaded model weights cached directly in `./models/` for 100% network isolation during ranking runs.
- **Low-Memory Streaming Parser:**
 $O(N)$ streaming JSONL parser ensures memory consumption stays under 2GB, preventing out-of-memory errors.
- **Portable OS Routing:**
Cross-platform path resolution supports Windows, Linux, and macOS out-of-the-box.

Results & Performance

COMPUTE BENCHMARKS	RANKING & SCREENING QUALITY	INTERACTIVE WEB APP RUN
■ Execution Speed: 193.75 seconds Completes full filtering, BM25 indexing, dense vector similarity, and Cross-Encoder re-ranking in 193.75s. ■ Memory Footprint: <2GB Peak RAM Filters 100k candidates line-by-line using under 2GB peak RAM, ensuring compatibility with standard 16GB CPU laptops. ■ Network Independence: 100% Offline Runs entirely offline with zero API calls, guaranteeing candidate data confidentiality.	■ Honeypot Screen Rate: 35,208 profiles Screened out 35,208 candidates with fraudulent skills or anomalous timelines, ensuring 0% honeypots in Top 100. ■ Monotonic Order compliance: Ensures scores strictly decrease as rank increases, with deterministic candidate_id sorting breaking duplicates. ■ Validation Suite: 100% Pass Checked and passed by validate_submission.py for strict schema, column ordering, and tie-breaker conformity.	■ JD Modulator & UI Shortlist: Allows real-time customization of Job Description and instantly compiles score meters (tech match, behavior). ■ Gradio Sandbox Server: Runs locally on port 7860, with web requests returning HTTP 200. ■ Sandbox Port Verification: Verified by urllib to successfully load and serve profile inspect tabs.

Evaluation & Ranking Quality

SHORTLIST STACK ACCURACY

- **1. Match Quality Alignment:**
Candidate profiles match the founding AI/ML engineer profile: RAG pipeline experience, vector databases, and evaluation metrics.

- **2. Title Match Calibration:**
Taxonomy mapping identifies candidates matching 'sde-iii', 'member of technical staff', or 'lead AI engineer' roles correctly.

- **3. Geographic Availability:**
Re-ranking incorporates candidate relocation readiness to filter out candidates bound to overseas or remote-only modes.

SHORTLIST COHERENCE & TRUST

- **1. Explanations Transparency:**
Programs generate transparent, non-generic descriptions (e.g. 'Software engineer with 6.9 years experience matching PyTorch, Milvus').

- **2. Factual Compliance:**
Zero hallucinated text or fabricated skills. Every claim is strictly backed by the candidate's career details.

- **3. Honeypot Screening Proof:**
Verified to contain 0% honeypots or timeline fraud candidates in the final recommended Top 100.

Submission Assets

GITHUB REPOSITORY

- **Clean Repository:**
Contains fully working, clean, and complete code with local model caches and Gradio app.
- **Repository Link:**
<https://github.com/anmolraj499-ai/Candidate-Ranking>
- **README Documentation:**
Detailed instructions on setup, environment, and pipeline execution.

REPRODUCIBILITY SANDBOX

- **Colab Notebook:**
Allows judges to run candidate evaluation sandbox on Google Colab's standard compute in seconds.
- **Sandbox Link:**
https://colab.research.google.com/github/anmolraj499-ai/Candidate-Ranking/blob/main/sandbox_ranking.ipynb
- **Automation Workflow:**
Clones the repo, installs dependencies, and runs ranker on sample_candidates.json.

SHORTLIST OUTPUTS

- **Validated Shortlist CSV:**
Format-compliant candidate rankings containing candidate_id, rank, score, and reasoning.
- **Excel Shortlist XLSX:**
Excel XLSX version created for easy upload through the portal's restricted format picker.
- **Submission Metadata YAML:**
Official metadata detailing compute environment, team contact details, and algorithms used.

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THANK YOU

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